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I have been reading a vast amount of popular science in the past month or so, nearly all brilliant, of course – what else would you expect New Scientist's culture editor to say? But it has also



been wonderful to take myself out of the real world with a slice of sci-fi from Martha Wells, **All Systems Red** (pictured).

The first in Wells's *The Murderbot Diaries* series, it introduces us to Murderbot, a part-robot, part-organic security unit that has overridden its programming and prefers to spend its time catching up on TV, while feeling obliged to protect the puny humans in its care. Murderbot's tone is brilliantly sardonic and this was a real escapist read for me. We are reading it in the New Scientist Book Club – do come and join us!

I have also been escaping to somewhere less pleasant with the **Fallout** TV series, which is set after a nuclear war and was highly recommended by our TV columnist, Bethan Ackerley. As a huge *Twin Peaks* fan, it is always damn fine to see Kyle MacLachlan in action.

Failure to launch

From flying cars to smart robots, why do many promised technologies remain out of reach, asks **Matthew Sparkes**



Book

The Long History of the Future

Nicole Kobie

Bloomsbury Sigma

(UK, on sale now; US, 24 September)

A HANDFUL of technologies teeter on the cusp of release but never arrive. Take driverless (or even flying) cars, superintelligent machines and human-like robots that free us from the drudgery of everyday chores. What stops them leaving the minds of inventors and entering our homes?

That is what Nicole Kobie asks in *The Long History of the Future: Why tomorrow's technology still isn't here*. While the answer can simply be because a given problem is very hard, she writes, it may also be that the proposed solution isn't very sensible and isn't at all the best way to solve the underlying issue.

Yes, flying cars would soar past congested traffic. But safe bicycle lanes would dissolve those queues. Yes, an underground network, or

hyperloop, of autonomous vehicles would whisk people from one end of a city to the other at high speeds. But so do some trains.

Kobie points out that many of these technological promises have longer and more complex histories than we realise. For instance, the term "artificial intelligence" was first coined in the 1950s, and the first "driverless" car was approved for sale in the US in 1956.

And robotics can be traced back to Leonardo da Vinci's "mechanical knight" in 1495, a fairly humble machine that could stand, sit and move its arms, but one that was a huge achievement for the time. So why, over 500 years later, are we still folding laundry ourselves?

As one academic tells Kobie in the book, a driverless car that works 95 per cent of the time is nowhere near good enough – you need 99.999 per cent. And every decimal place of reliability takes another five to 10 years to achieve. The failure of robots, driverless cars and other devices to materialise in a useful form isn't through lack of effort or investment, it is that they are only useful once they are useful, and that is a very hard problem to solve.

Kobie does a wonderful job of teasing out key moments

and concepts, weaving together a compelling history of long-awaited technologies. It is easier to understand why, say, driverless cars haven't arrived once you know researchers have been working on them ever since cars were invented.

The string of failed attempts, dead-ends and premature launches are as entertaining as they are cautionary. One such example was the 1947 ConvAirCar, which could work as both car and plane, to some extent. On a test flight, the pilot crashed when he looked at the wrong dashboard – there were two, one for the plane, one for the car. He didn't notice he had run out of fuel.

Another problem Kobie sees with technological progress is the cost, often needing the sort of talent and cash available in Silicon Valley. This tends to place the power of deciding what gets built in the hands of a few powerful white men – people who may be insulated from the everyday problems that really need solving.

But for all Kobie's perfectly valid points about the slow growth of technology in some areas and the misguided efforts of some really smart people with enviable funding, it is worth pointing out that science still throws up unexpected delights.

Who could have foreseen just five years ago that we could use AI to predict the structure of nearly every protein known to science? Or that "transformer" models (despite their faults) would create coherent essays, photorealistic images and even whole songs?

And there are other long-predicted technologies that now seem genuinely close to roll-out. Within a few years, we could have useful quantum computers, and fusion power is really edging closer.

Kobie is right that some problems are far bigger and thornier than we thought, but we should never forget the power of the unexpected along the enchanted route to progress. ■

